



# Using Bayer Pattern to Detect Fake Images

Larry Liu, Ziyao Wang, Eric Cao, Olasunade Ladenegan

February 2024

---

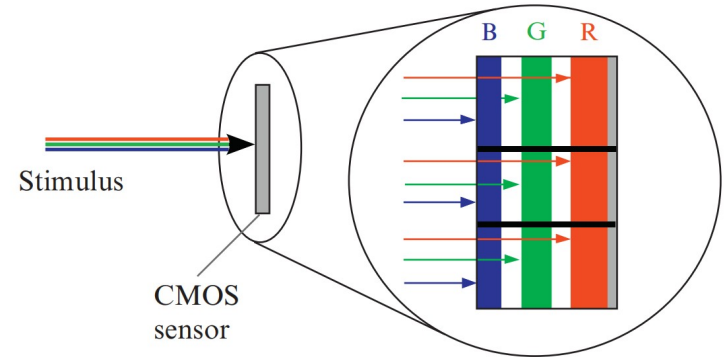
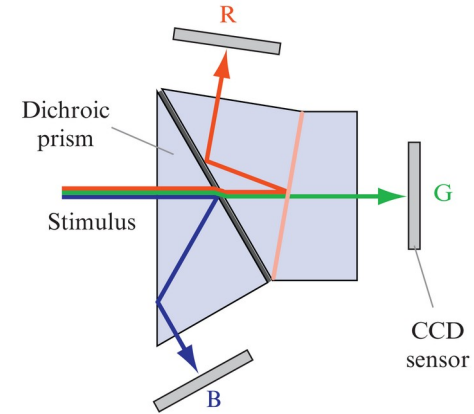
# Synthesized Images

- Even human eyes may fail to recognize images synthesized by deep learning.
- Many deep generative algorithms are available:
  - Generative adversarial networks
  - Variational autoencoder
  - Diffusion and score-based models
  - ...



# Cameras and Photosensors

- CMOS and CCD photosensors convert light intensity to analog signals.
- Can't record intensity of different wavelength.
- For an image to be colorful, at least three color components are needed at each pixel position.



# Cameras and Photosensors

- Cameras with three sensors are structurally complex and expensive.
- Cover a color filter array over the sensor to capture one primary color component at each location.
- Interpolate the missing colors in the post-processing pipeline.

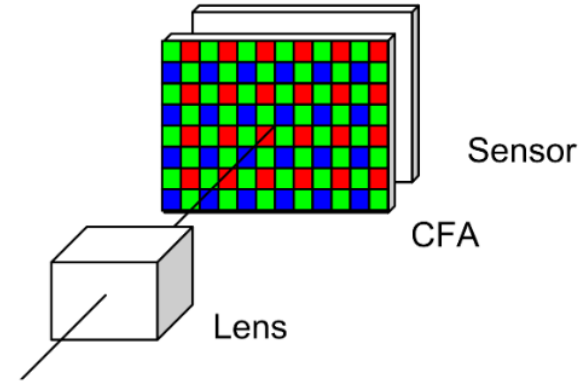


Image source: Xin Li, Bahadır Gunturk, and Lei Zhang. Image demosaicing: A systematic survey. In Visual Communications and Image Processing 2008, volume 6822, pages 489–503. SPIE, 2008.

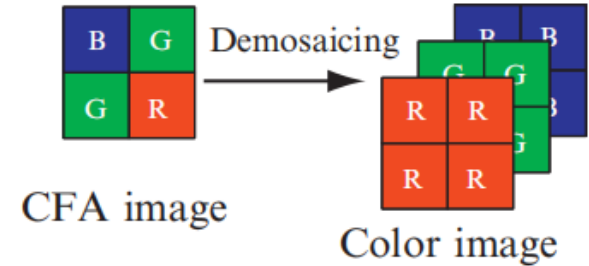
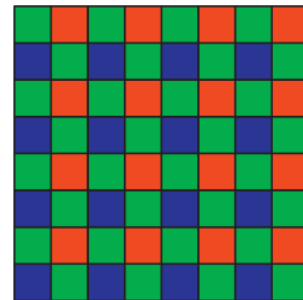


Image source: Olivier Losson, Ludovic Macaire, and Yanqin Yang. Comparison of color demosaicing methods. In Advances in Imaging and Electron Physics, volume 162, pages 173–265. Elsevier, 2010.

# Bayer Filter

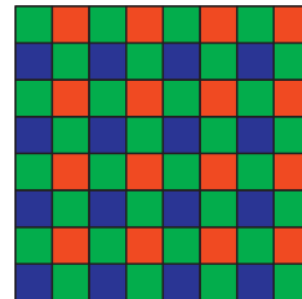
- Introduced by Bryce Bayer in the Kodak company in 1976
- Cone cells in human eyes can better resolve green in daylight vision



---

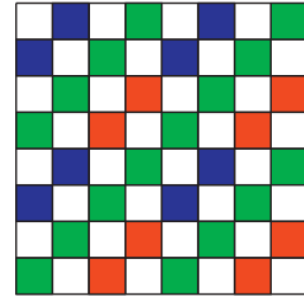
# Bayer Filter

- Introduced by Bryce Bayer in the Kodak company in 1976
- Cone cells in human eyes can better resolve green in daylight vision



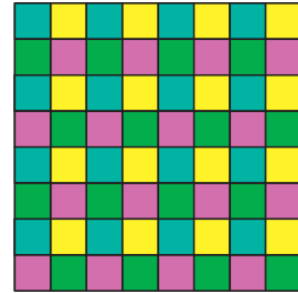
# Kodak's CFA 2.0

- Featuring panchromatic filters that is transparent to all light.
- Can capture a greater range of light even in dark environments.
- Less accurate on color.



# CYM-Based Filters

- Use cyan, magenta, and yellow as primary color components.
- Less light-absorbing blocks.
- CYM is a subtractive color space. RGB is an additive color space. Interpolating the missing colors at each position is trickier.

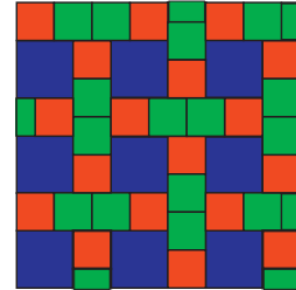




---

# Burtoni Filter

- Proposed by **Ronald H H Kroger**
- Drew inspiration from the cone cells of cichlid fish
- Reduce aliasing artifacts by incorporating more spatial frequencies and geometris of different colors
- Trickier to demosaic



# Demosaicing of Bayer Patterned Color Images

- Digital cameras use charge-coupled device (CCD) sensor, with the CCD pixels preceded in the optical path by a color filter array (CFA) in a Bayer mosaic pattern
- Demosaicing: the problem of interpolating back the images captured by CFA

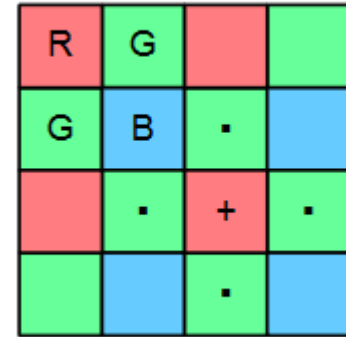


Image source: Henrique S Malvar, Li-wei He, and Ross Cutler. High-quality linear interpolation for demosaicing of bayer-patterned color images. In

2004 IEEE International Conference on Acoustics, Speech, and Signal Processing, volume 3, pages iii–485. IEEE, 2004.

# Bilinear Interpolation

At pixel position  $(i,j)$ , estimating the green value:

$$\hat{g}(i,j) = \frac{1}{4} \sum_{(m,n) \in \{(0,-1), (0,1), (-1,0), (1,0)\}} g(i+m, j+n)$$

|   |   |   |   |
|---|---|---|---|
| R | G |   |   |
| G | B | • |   |
|   | • | + | • |
|   |   | • |   |

# Bilinear Interpolation

Pros:

Simple to compute

Cons:

Generates significant artifacts, especially across edges and other high-frequency content

|   |   |   |   |
|---|---|---|---|
| R | G |   |   |
| G | B | • |   |
|   | • | + | • |
|   |   | • |   |

# Gradient-corrected bilinear interpolation

Edges have much stronger luminance than chrominance components

There's a sharp luminance change at that pixel when there's a sharp change in one color channel

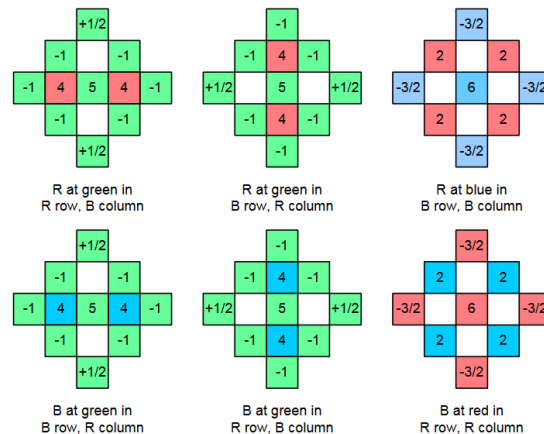
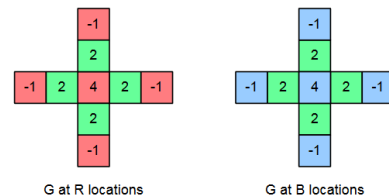
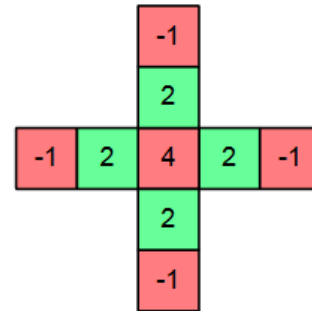


Image source: Henrique S Malvar, Li-wei He, and Ross Cutler. High-quality linear interpolation for demosaicing of bayer-patterned color images. In 2004 IEEE International Conference on Acoustics, Speech, and Signal Processing, volume 3, pages iii–485. IEEE, 2004.

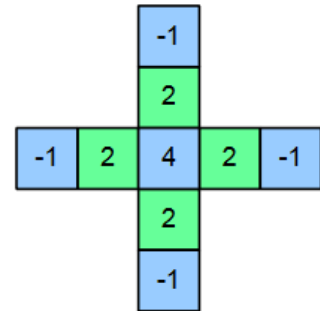
# Gradient-corrected bilinear interpolation

$$\hat{g}(i, j) = \hat{g}_B(i, j) + \alpha \Delta_R(i, j)$$

$$\Delta_R(i, j) \triangleq r(i, j) - \frac{1}{4} \sum_{(m,n) \in \{(0,-2), (0,2), (-2,0), (2,0)\}} r(i+m, j+n)$$



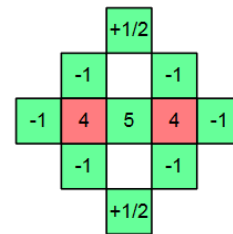
G at R locations



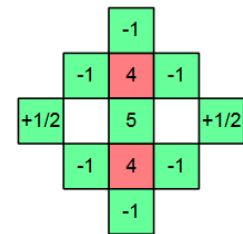
G at B locations

# Gradient-corrected bilinear interpolation

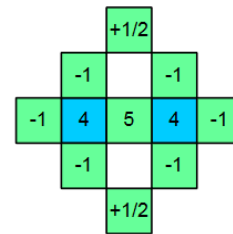
$$\hat{r}(i, j) = \hat{r}_B(i, j) + \beta \Delta_G(i, j)$$



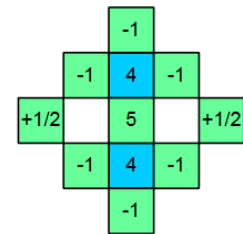
R at green in  
R row, B column



R at green in  
B row, R column



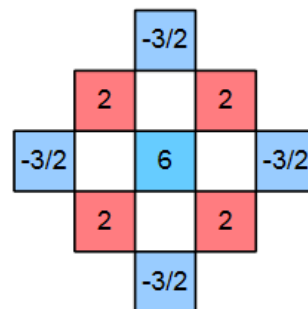
B at green in  
B row, R column



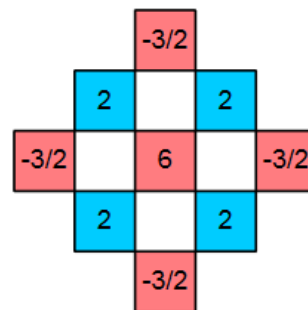
B at green in  
R row, B column

## Gradient-corrected bilinear interpol

$$\hat{r}(i, j) = \hat{r}_B(i, j) + \gamma \Delta_B(i, j)$$



R at blue in  
B row, B column



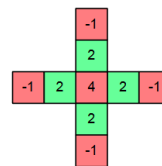
B at red in  
R row, R column



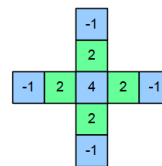
# Gradient-corrected bilinear interpolation

Gain factor:

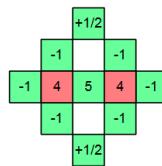
$$\alpha = 1/2, \beta = 5/8, \text{ and } \gamma = 3/4$$



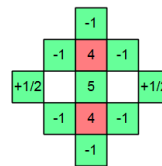
G at R locations



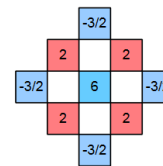
G at B locations



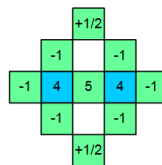
R at green in  
B row, B column



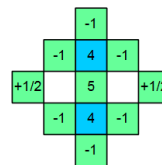
R at green in  
B row, B column



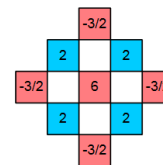
R at blue in  
B row, B column



B at green in  
B row, R column



B at green in  
R row, B column

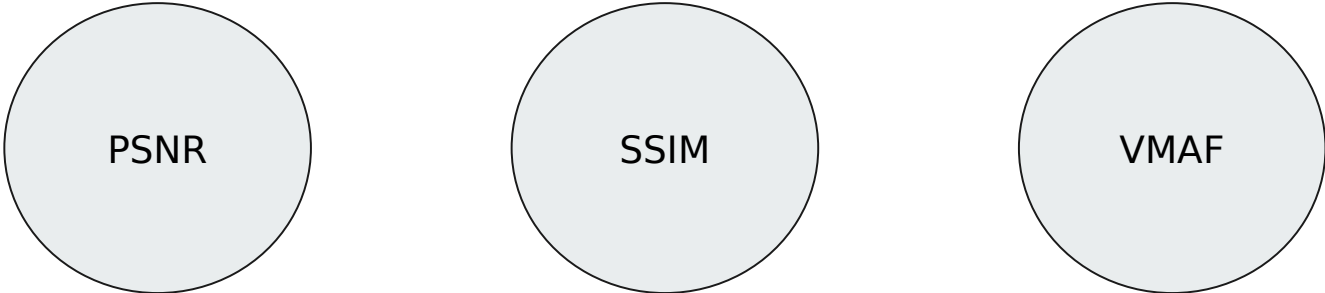


B at red in  
R row, R column

---

# Metrics for Evaluation

The three primary metrics that would be used for evaluation of our hypothesis are



PSNR

SSIM

VMAF

# Peak Signal-to-Noise Ratio (PSNR)

- The peak signal to noise ratio is the ratio of the maximum possible signal power to the power of the distorting noise that affects the quality of its representation.
- PSNR values are measured in decibels (dB).
- The higher the PSNR value, the better the image quality.

$$MSE = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

The PSNR (in dB) is defined as:

$$\begin{aligned} PSNR &= 10 \cdot \log_{10} \left( \frac{MAX_I^2}{MSE} \right) \\ &= 20 \cdot \log_{10} \left( \frac{MAX_I}{\sqrt{MSE}} \right) \\ &= 20 \cdot \log_{10} (MAX_I) - 10 \cdot \log_{10} (MSE) \end{aligned}$$

# Structural Similarity Index Measure (SSIM)

- The SSIM metric is a perception based model that uses structural information to measure the quality of images
- It breaks image distortion into the 3 components below to quantify image quality.
  - Luminance
  - Contrast
  - Structure
- SSIM values are on a scale of 0 -1
- It aligns well with the Human Visual System and has great accuracy.

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}.$$

## Structural Similarity Index Measure (SSIM)

$$\mu_x = \sum_{i=1}^N w_i x_i$$

$$\sigma_x = \left( \sum_{i=1}^N w_i (x_i - \mu_x)^2 \right)^{\frac{1}{2}}$$

$$\sigma_{xy} = \sum_{i=1}^N w_i (x_i - \mu_x)(y_i - \mu_y).$$

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = [l(\mathbf{x}, \mathbf{y})]^\alpha [c(\mathbf{x}, \mathbf{y})]^\beta [s(\mathbf{x}, \mathbf{y})]^\gamma,$$

$$l(\mathbf{x}, \mathbf{y}) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2\mu_y^2 + C_1},$$

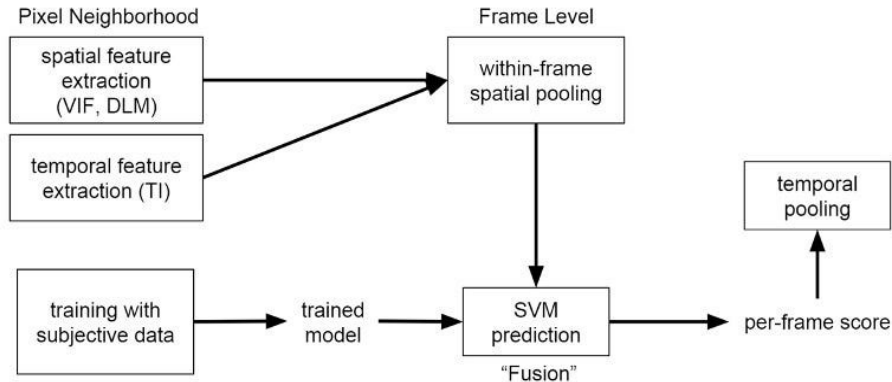
$$c(\mathbf{x}, \mathbf{y}) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2\sigma_y^2 + C_2},$$

$$s(\mathbf{x}, \mathbf{y}) = \frac{2\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3}.$$

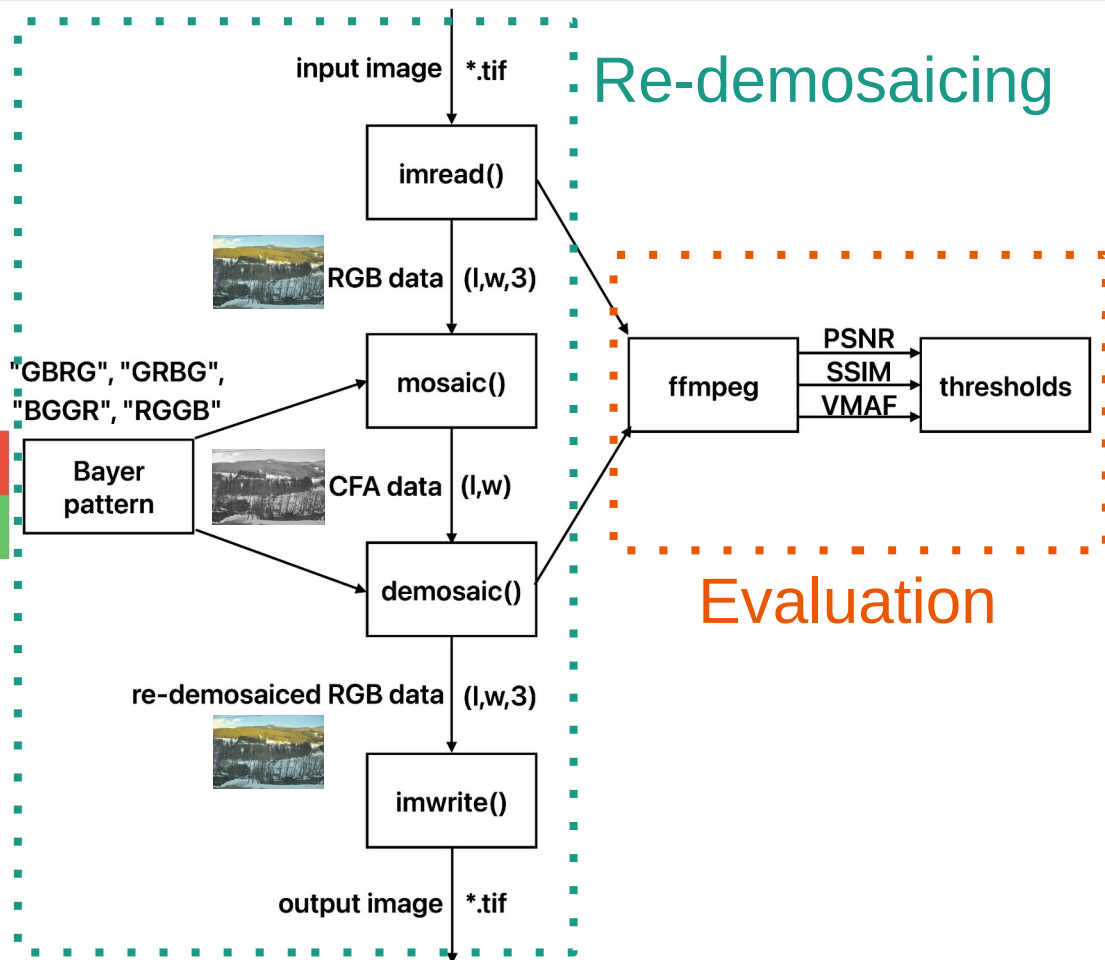
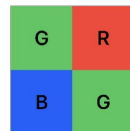
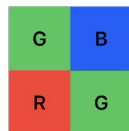
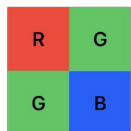
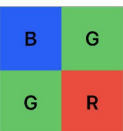
# Video Multimethod Assessment Fusion (VMAF)

- VMAF is a metric that was developed by Netflix and 3 universities
- VMAF combines multiple image quality metrics using machine learning to predict image quality.
- VMAF scores range from 0-100.

## How VMAF works



# Implementation



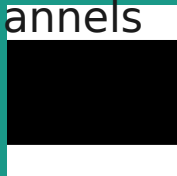
# Mosaic(): RGB -> CFA

5x5x3 RGB  
Image



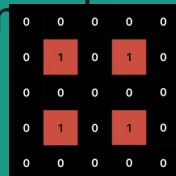
RGB  
channels

R



×

BGGR  
mask

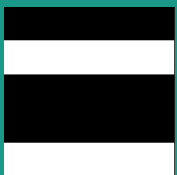


=

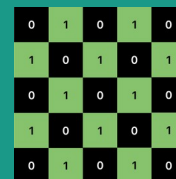
selected  
RGB



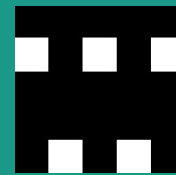
G



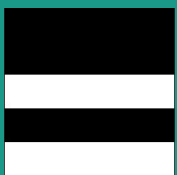
×



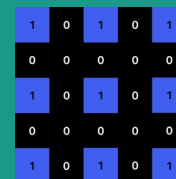
=



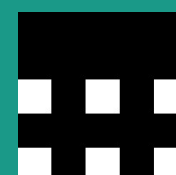
B



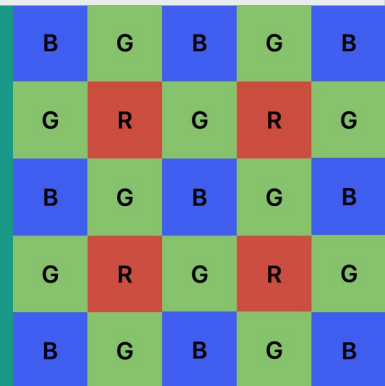
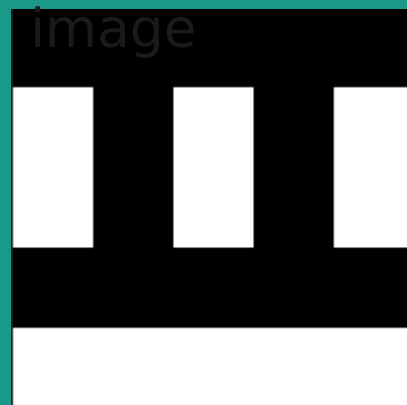
×



=



5x5 CFA  
image

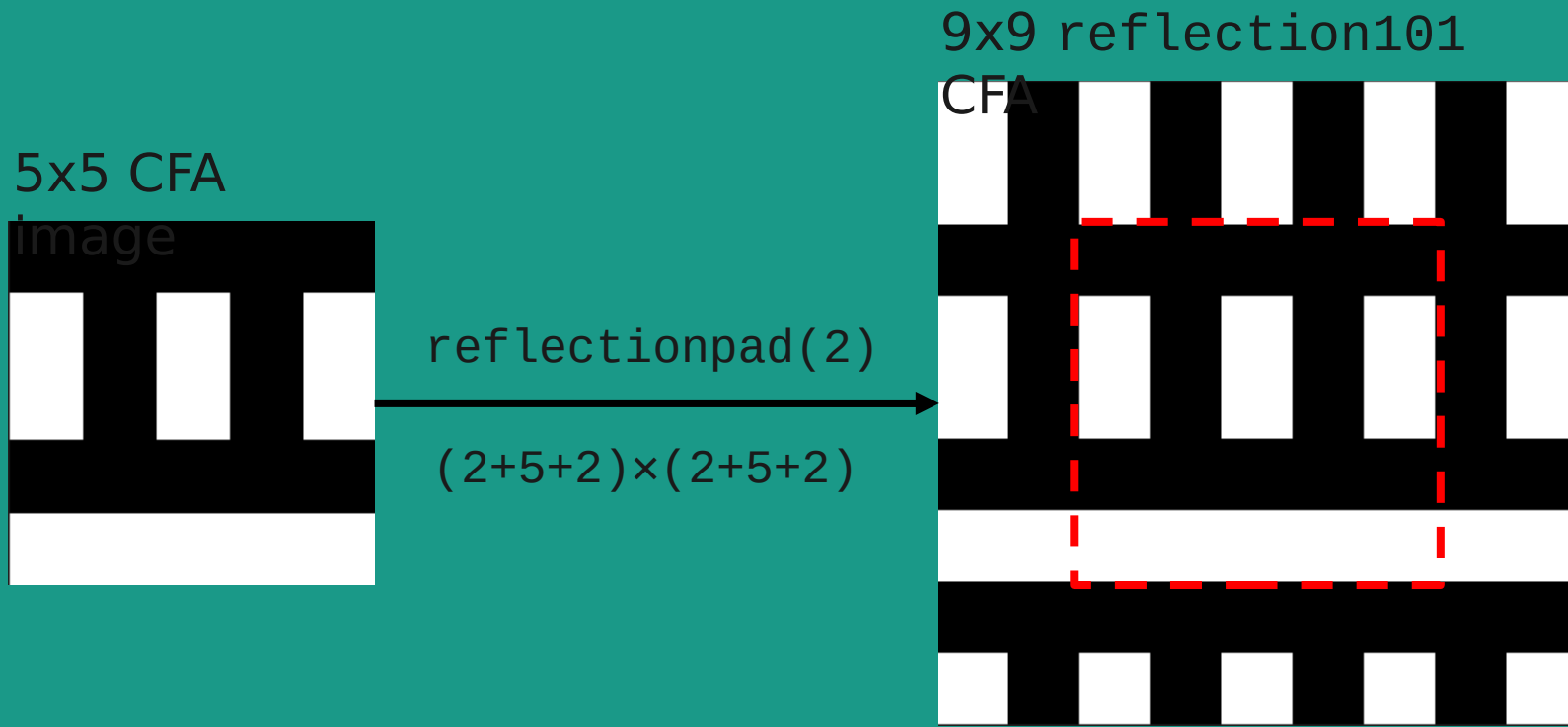




## Demosaic(): CFA -> RGB

- Gradient-corrected bilinear interpolation [*Malvar et. al. 2004*]
- 5x5 convolution kernels to correct bilinear interpolation
  - `OpenCV.filter2D()`
  - `PyTorch.conv2d()`

## PyTorch.conv2d(): corrected interpolation



```
conv2d(9x9_reflection_CFA, 5x5_GR_GB_kernel)
```

5x5 GR\_GB  
kernel

|       |     |       |     |       |
|-------|-----|-------|-----|-------|
| 0     | 0   | -.125 | 0   | 0     |
| 0     | 0   | .25   | 0   | 0     |
| -.125 | .25 | .5    | .25 | -.125 |
| 0     | 0   | .25   | 0   | 0     |
| 0     | 0   | -.125 | 0   | 0     |

|       |     |       |     |       |     |       |     |       |
|-------|-----|-------|-----|-------|-----|-------|-----|-------|
| 0     | 0   | 0     | 0   | 0     | 0   | -.125 | 0   | 0     |
| 0     | 0   | 0     | 0   | 0     | 0   | .25   | 0   | 0     |
| -.125 | .25 | -.125 | .25 | -.125 | .25 | .5    | .25 | -.125 |
| 0     | 0   | 0     | 0   | 0     | 0   | .25   | 0   | 0     |
| 0     | 0   | 0     | 0   | 0     | 0   | -.125 | 0   | 0     |

5x5 G  
channel

channel

## **PSNR, SSIM, VMAF: poor similarity -> fake image**

- measure similarity between original and re-demosaiced image
- normal distribution, quantiles as thresholds
- logistic regression, learn thresholds

